

Boundless Conversations: AI-Powered Video Interactions across Domains, Languages, and Time

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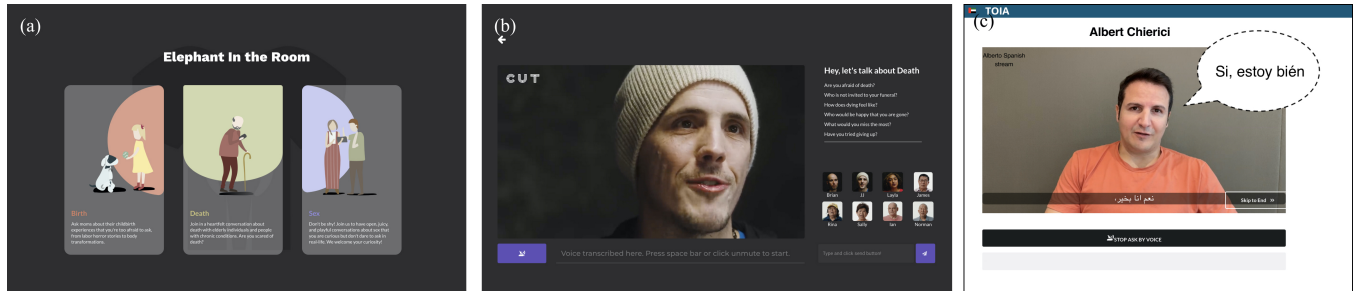


Figure 1: (a) *Elephant in the Room*'s home screen, enabling topic selection; (b) Video screen during an end-of-life conversation; (c) Multilingual interaction, showcasing a user questioning in Arabic and receiving a Spanish response with Arabic subtitles.

ABSTRACT

We present novel web applications that blend AI and video communication, offering an interactive way to engage with videos, transcending time and distance constraints. Known as Time Offset Interaction Application (TOIA), the platform transforms passive video viewing into dynamic conversations using personal recordings or YouTube clips. Our work expands TOIA with two unique experiences: 1) "The Elephant in the Room" project, stimulating dialogues on sensitive topics like childbirth, sex, and death; 2) Support for multilingual interactions, fostering a global collection of diverse personal narratives. Through AI and interactive media, we aim to broaden the horizons for sharing human experiences and perspectives.

ACM Reference Format:

Alberto Chierici, Soojin Lee, Muhammad Ali, Gautham Dinesh Kumar, Bishnu Dev, Nizar Habash, and Aaron Sherwood. 2023. Boundless Conversations: AI-Powered Video Interactions across Domains, Languages, and Time. In *SIGGRAPH Asia 2023 Emerging Technologies (SA Emerging Technologies '23)*, December 12–15, 2023, Sydney, NSW, Australia. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3610541.3614572>

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SA Emerging Technologies '23, December 12–15, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0312-6/23/12.

<https://doi.org/10.1145/3610541.3614572>

1 INTRODUCTION

Conversations in the real world encompass diverse forms, varying in authenticity, consistency, depth, directionality (uni- or bi-directional), length, and the impact they leave on participants. Time Offset Interaction Application (TOIA) emerges as a technology contributing to asynchronous human interaction [Abu Ali et al. 2018; Chierici et al. 2021; Chierici 2023; Traum et al. 2015]. TOIA reshapes the conventional static nature of video communication by facilitating the recording of video responses to queries and catalyzing genuine dialogues.

Currently uni-directional, TOIAs embody intrinsic attributes akin to genuine conversations: depth, meaningfulness, and enduring impressions. Their exceptional strength resides in engaging broad audiences simultaneously, surmounting time and language barriers. Unlike passive video consumption, TOIA empowers participants to guide discussions, heightening psychological engagement. This comes with trade-offs, e.g., using deep fake technology could make conversations bi-directional but lose authenticity.

This paper explores TOIA's potential by presenting two innovative extensions: 1) The *Elephant in the Room* project (EitR), an art installation fostering engagement with intimate subjects such as childbirth, sex, and death; 2) Enhanced multilingual experiences. Through these facets, we spotlight TOIA's capacity for dynamic video content interaction, delving into its technical intricacies and user encounters.

2 SYSTEM OVERVIEW

TOIA provides an accessible interface for TOIA-avatar creation and interaction, where users record video responses to various questions. These responses are saved and available for exchange through voice or text input or question cards.

At the time of writing, OpenAI GPT-3 models power the TOIA Dialogue Manager (DM), which employs semantic text searches, converting user queries and associated video text into vector embeddings [Neelakantan et al. 2022].¹ It identifies the most suitable video response using cosine similarities between the vector embedding of a new query and the question-answer pairs in the database. As such, the system can find answers for any free-form question, but if the DM does not find a strong enough match (cosine similarity falls below a certain threshold), it selects a random video between recordings made for this scenario (e.g., “Sorry, I haven’t recorded an answer for this.”), ensuring a sensible and timely response.

We expanded the scope and technological development of the existing TOIA for EitR. This interactive media art project provides visitors an intimate space to address the ‘elephant in the room’: engage with sensitive yet important topics people are often uncomfortable discussing. We achieved this result by integrating interview-like videos into the system (e.g., from YouTube) and making them interactive. Creating such experiences involves video scraping, trimming, uploading, and metadata tagging, thus transforming conventional videos into clips of interactive TOIA-avatars. For instance, a video clip where a character talks about a specific experience would be tagged with a related question, allowing users to engage in interactive discussions on deep topics.

We further extend the TOIA’s possible experiences by introducing multilingual capabilities, going beyond the limited TOIA Arabic-English bilingual prototype by [Abu Ali et al. 2018]. Our approach dynamically adjusts language based on user preferences across TOIA modules. TOIA-avatar creators can record videos in English, Arabic, Spanish, or French, enriching their stories, and we keep adding more languages. These recordings are processed in the background by a translation micro-service, which produces transcript translations and subtitle files. At the same time, a microservice accessible by API generates multilingual semantic vectors (embeddings) in real time, and we extend the DM for computing similarity scores between multilingual embeddings for optimal answer selection. The model we chose for the task was the Language Agnostic BERT Sentence Embeddings (LaBSE) [Feng et al. 2020], which spans over 100 languages. The interactive interface allows conversations in multiple languages, with video answer retrieval and subtitles in the user’s chosen language.

3 USER EXPERIENCE

Given the sensitive nature of EitR, we envision our system demonstration to be installed in a booth or analogous space that can guarantee a private conversation. To ensure visitors’ comfort, it is appropriate to display a content warning banner informing visitors of potentially triggering content. Additionally, a data policy should be displayed indicating that no personal data (video, audio,

or transcript) will be stored or retained after the conference. Both use cases of TOIA will be available on the same computer.

3.1 Elephant in the Room

Visitors interact with EitR by using a laptop and accessing it by clicking on the appropriate link. One can then select a topic (Figure 1(a)) like death. The interaction page allows for voice or text input for questions, with prompts available but not compulsory (Figure 1(b)). They serve as ice-breakers. The system retrieves an appropriate video response when a user inquires, for instance, “Do you think anyone would be happy that you are dead?”. The interface presents video answers to one topic from different characters. Visitors can address specific characters, such as “JJ, what do you miss the most?”

3.2 Multilingual Avatar Creation & Interaction

Visitors can access the multilingual TOIA by going to the appropriate link (Figure 1(c)): they can interact with and create TOIA-avatars. The system already has four pre-loaded TOIA-avatars representing two individuals, Persona A (English and Spanish) and Persona B (English and Arabic), each with similar questions in their databases but unique answers. These are university students who provided responses to about 60 questions about themselves and their lives on campus. One can interact with the English and Spanish/Arabic TOIA-avatars, asking them questions in English, Spanish, Arabic, or French (and we may add more languages over time). Multilingual support encompasses source language subtitles and corresponding questions; dubbed videos or versions are unnecessary. By creating a new account, visitors can also create their own TOIA-avatar and decide if making them available for other guests to interact with or to keep private and play with multilingual interaction with oneself.

4 CONCLUSION AND FUTURE WORK

We presented two novel web applications that blend AI and video communication within the TOIA open-source framework (GitHub, Demo). Future work includes extensions to other applications and continually improving the framework’s accuracy and efficiency.

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¹We chose GPT-3 for user-experience studies we ran in 2022. The DM is modular, and we keep testing and updating it with newer techniques and language models.